

Lecture 2: Plate Tectonics

Background(1/2)

- There is tangible evidence chiefly from palaeomagnetism that the crust (oceanic and continental) together with the upper part of the mantle which overlie the asthenosphere constitute the lithosphere, which is disjointed into plates or blocks by faults or thrusts.
- Thus the lithosphere is made up of lithospheric plates.

Background(2/2)

- Plate tectonic provide a modern view of the rock cycle.
- It involves a worldwide network of moving plates of the lithosphere.

Important features of plate tectonics

- It is assumed that the earth is composed of 20 lithospheric plates.
- They have a thickness of 0-10 km at the ridge to 100 -150 km elsewhere.
- The plates may contain continental as well as oceanic surfaces
- The plates are continuously in motion both with respect to each other and to the earth's axes of rotation.

Important features of plate tectonics

- Virtually all seismicity, volcanic activity is localized around plate margins and is associated with differential motion between adjacent plates.
- They move with velocities ranging from 1 – 6 cm per year.
- Where two plates diverge we find extensional features typically oceanic ridges symmetrical about the vertical axis. – **constructive plate margin**



Important features of plate tectonics

- where two plates converge and one is thrust beneath the other find island arcs. These are sites of great earthquakes, explosive volcanism great topographic relief and other features – **destructive plate margin**
- slide past each other we find transcurrent faults. – **conservative plate margin** (no gain or loss of surface area)

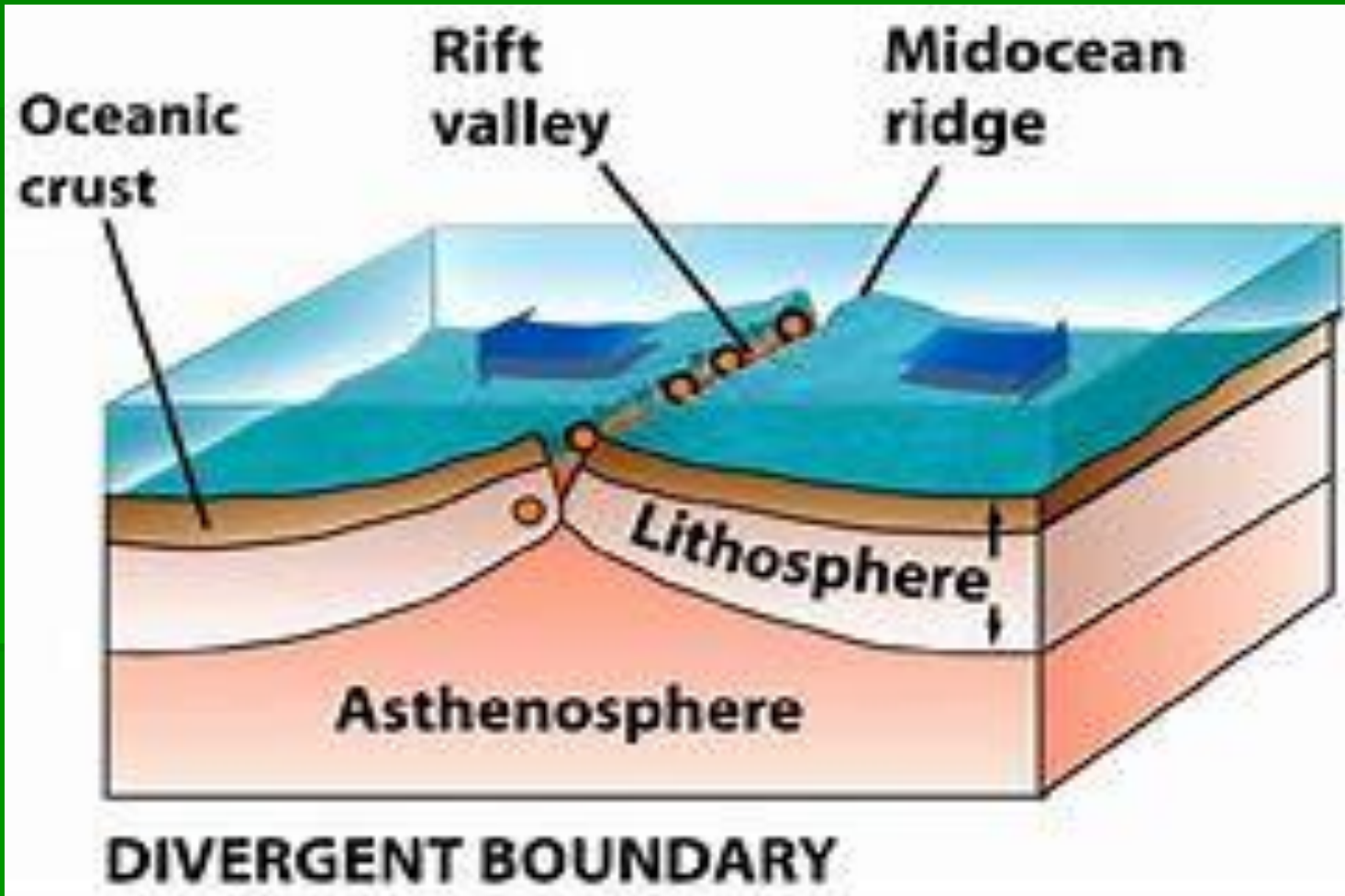
Type of plate-margin -3 types

- **Constructive** – new crust is created by upwelling of materials from the mantle.
- The lithospheric plates diverge at the crest of the mid oceanic ridges where new surface is created. Plates move away from each other but do not separate. “sources”

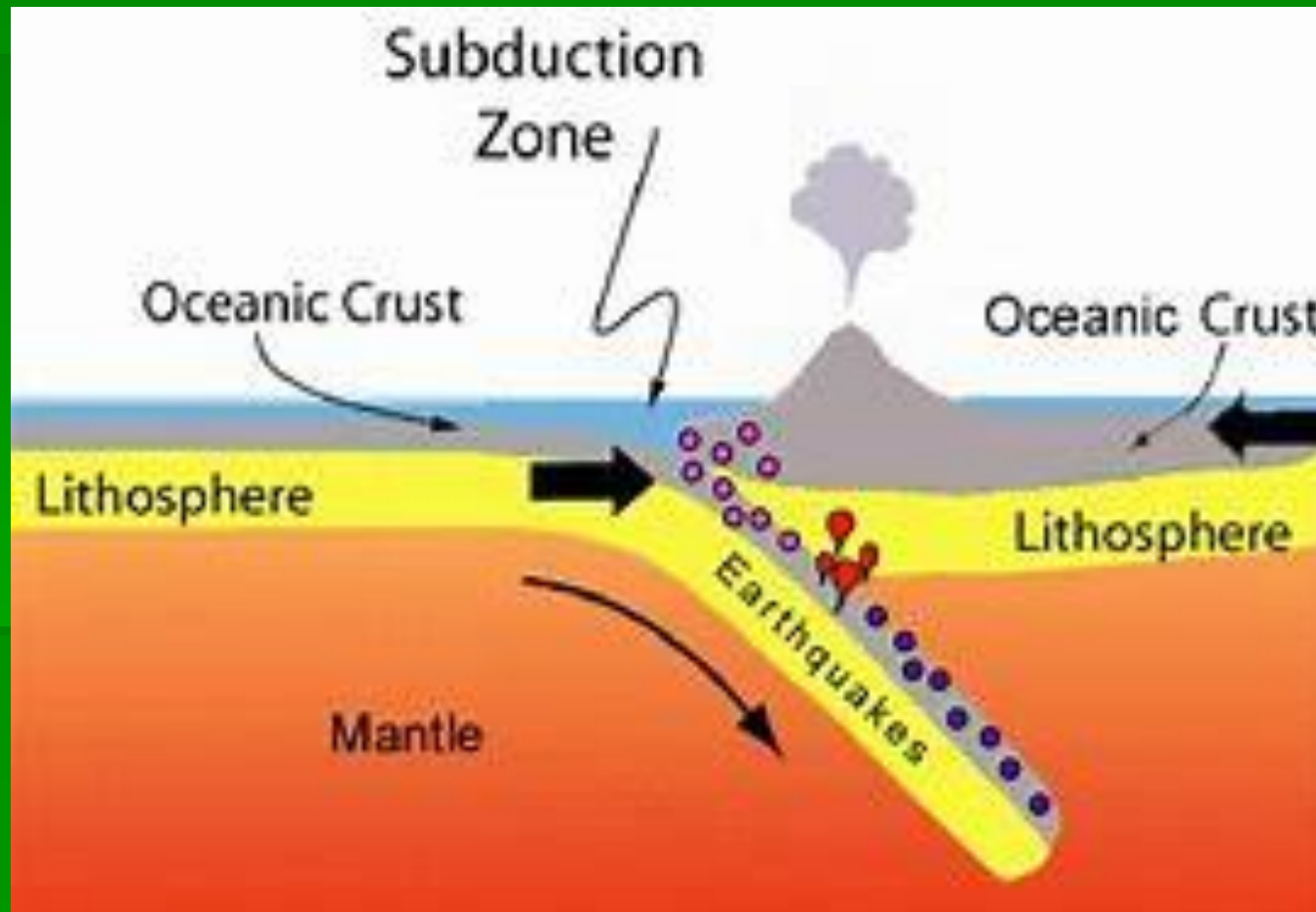
Type of plate-margin -3 types

- **Destructive** – a zone of convergence of plate boundaries.
- Two lithospheric plates are coming together and one plate is forced to plunge down into the mantle.
- The plate boundary is a reverse fault dipping at about 30 degrees from the horizontal
- The overriding of one plate on another gives rise to trenches and island arcs.
„sinks“

Constructive plate margin



Destructive plate margin(1/2)



Destructive plate margin(2/2)

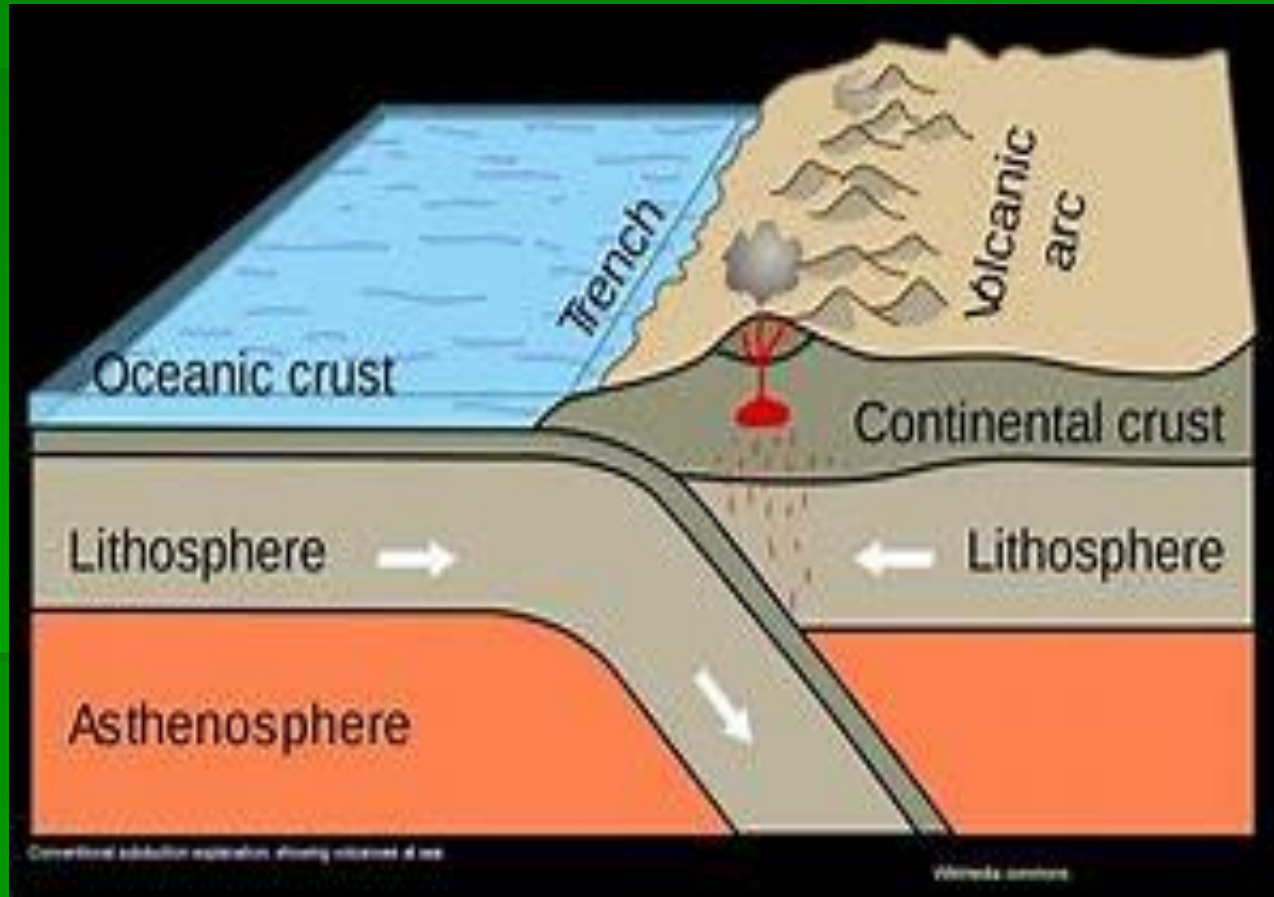


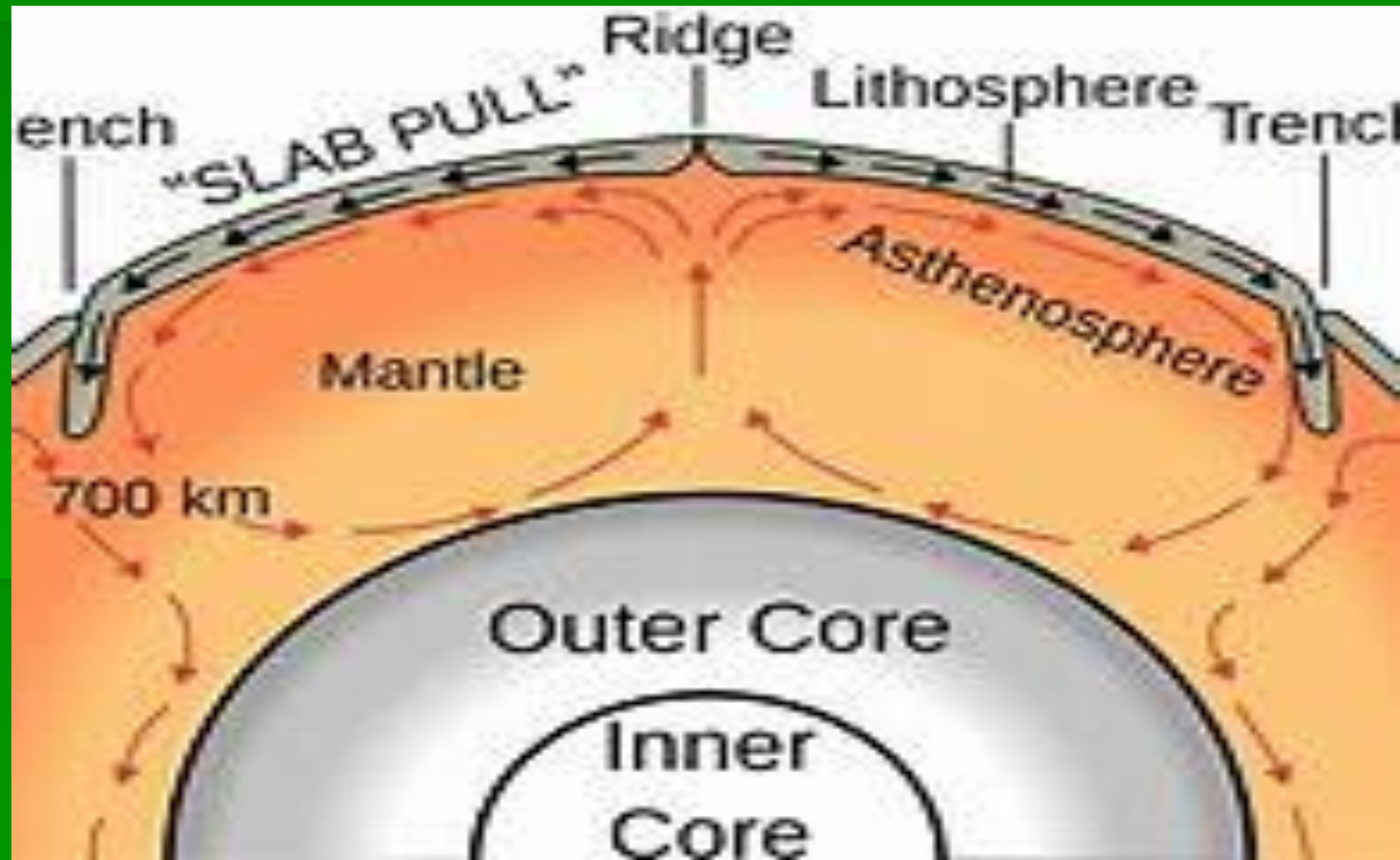
Plate margins marked by earthquakes and volcanoes



Type of plate-margin -3 types

- **Conservative** – where plates can slide past each other and plates neither gain or lose surface area there results a trancurrent or transform fault.

Interaction between mantle and lithosphere



Importance of plate tectonics

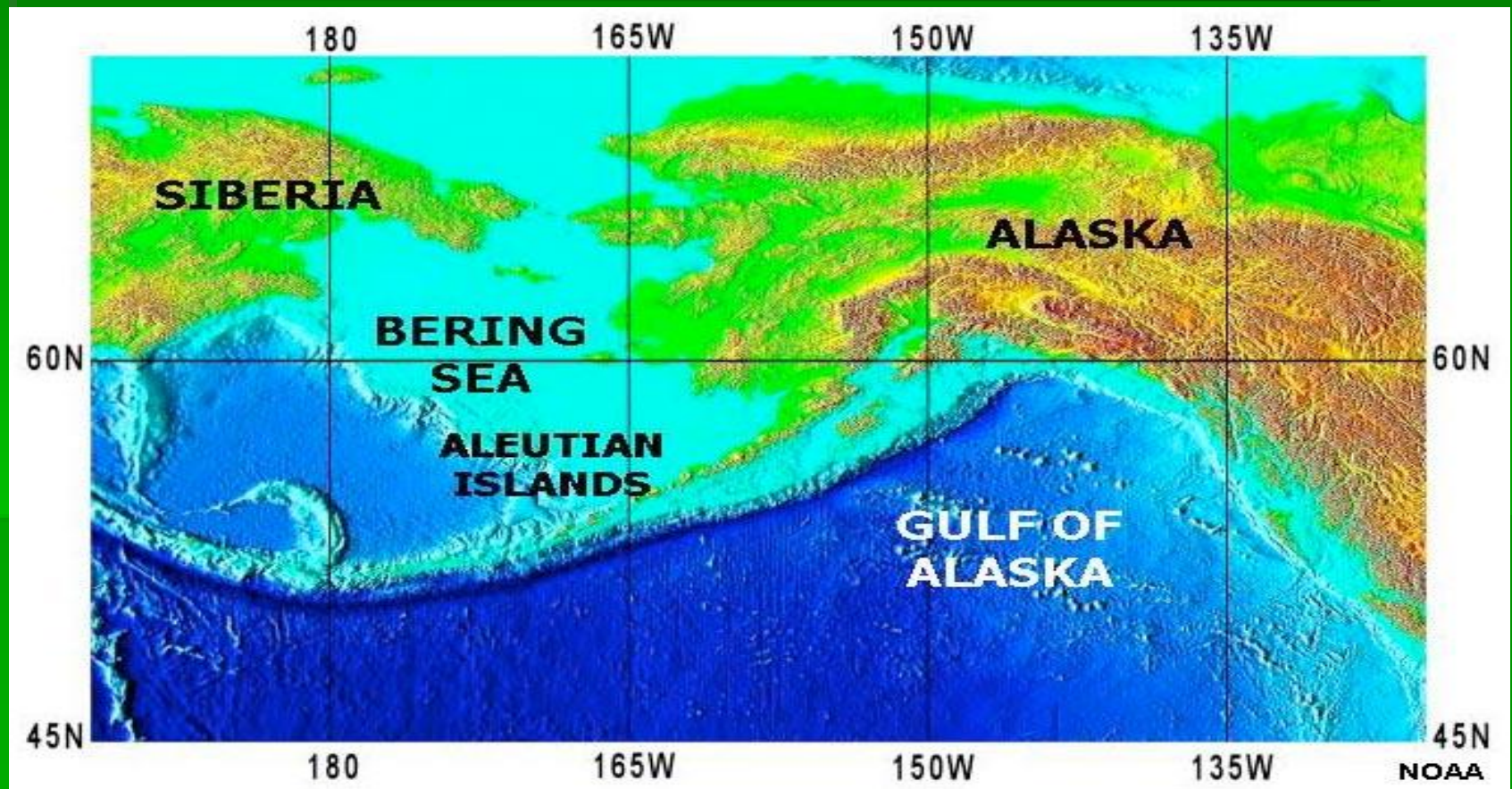
- The theory of plate tectonics is useful in explaining:

- Continental drift
- Mountain building
- Island arcs oceanic trenches
- Ocean flow spreading
- Mid-oceanic ridges
- Pattern of seismicity that we see.

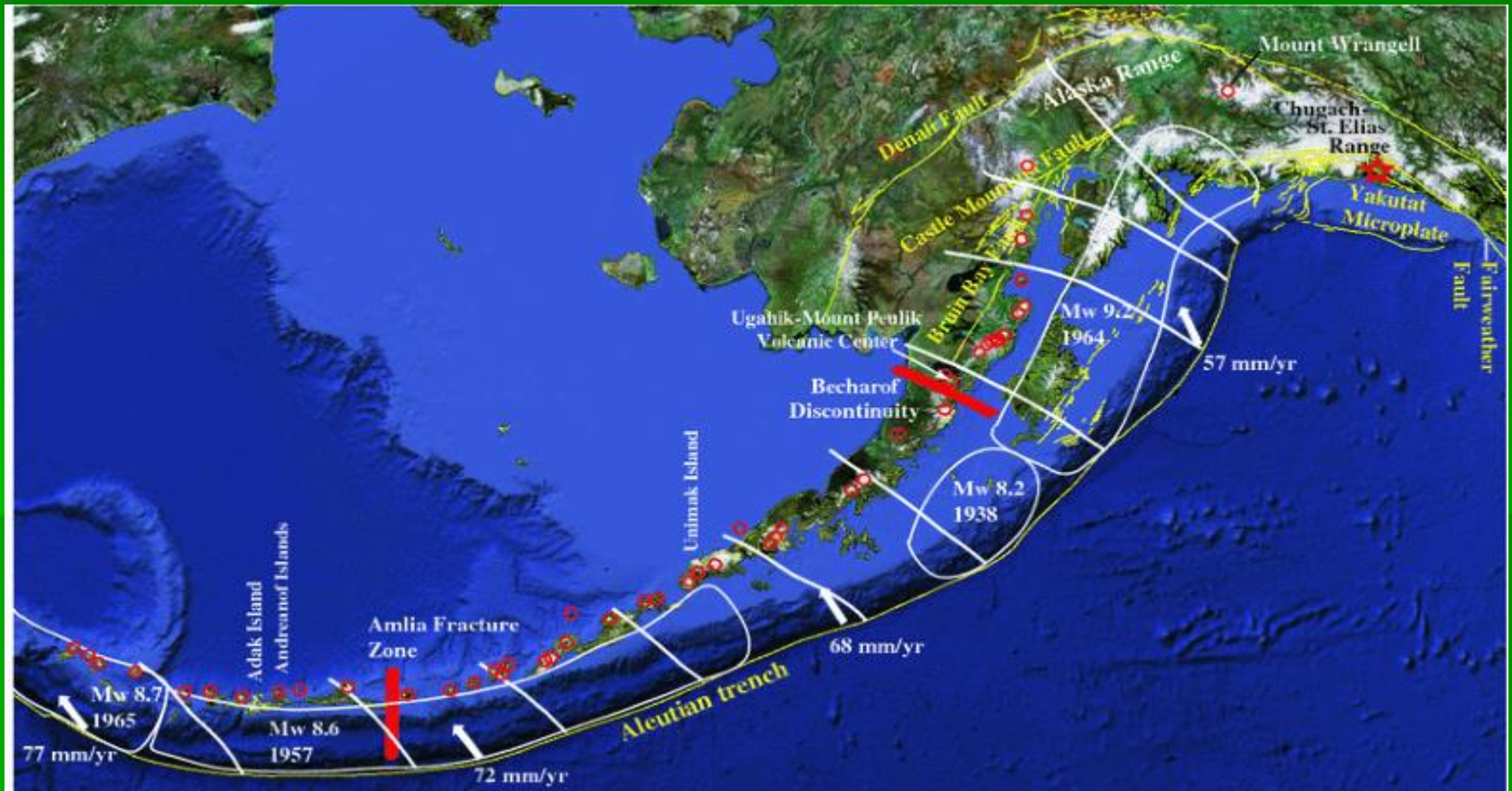
Evidence in support of plate tectonics

- Shapes of continents – match of coastlines
- Pattern of location of earthquakes and volcanoes
- Sea floor magnetization at mid-oceanic ridges
- Continuity of location of mountains and fossils when continents are fitted together
- Direct measurements (GPS)

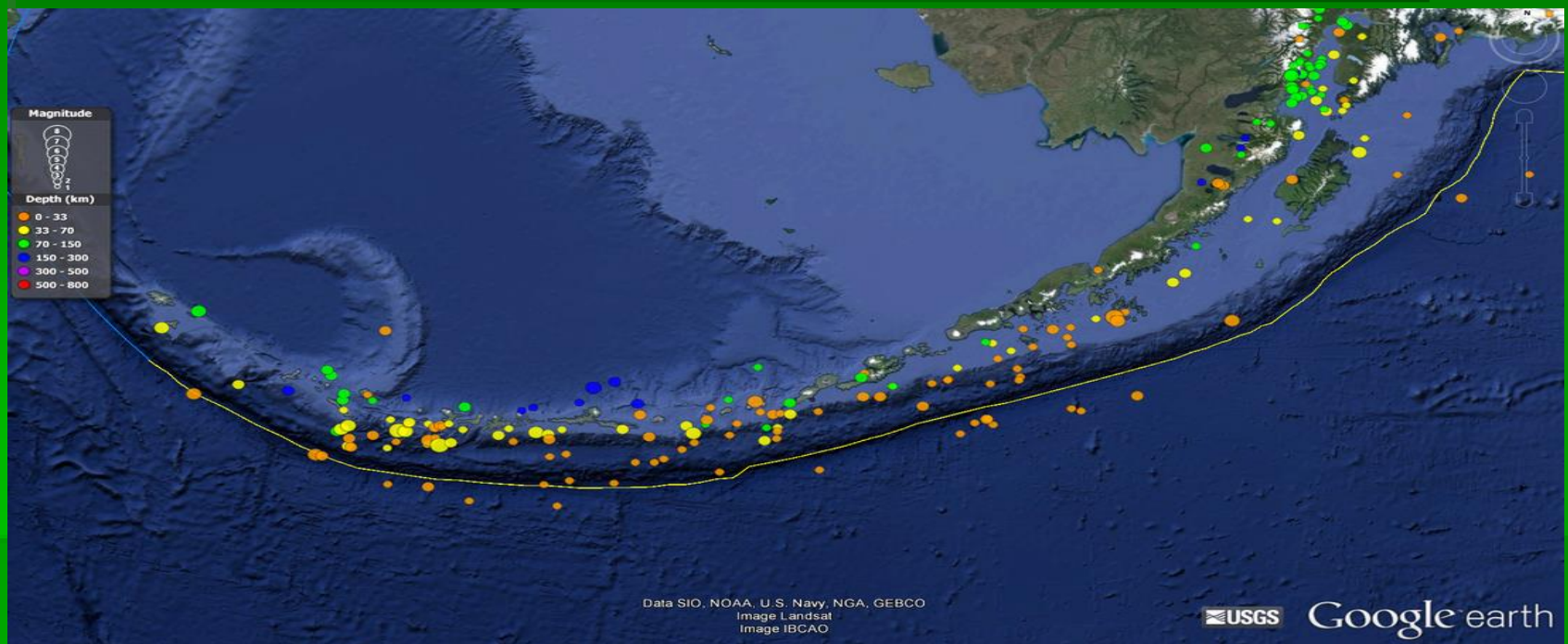
Ocean continent collision



Ocean continent collision



Ocean continent collision

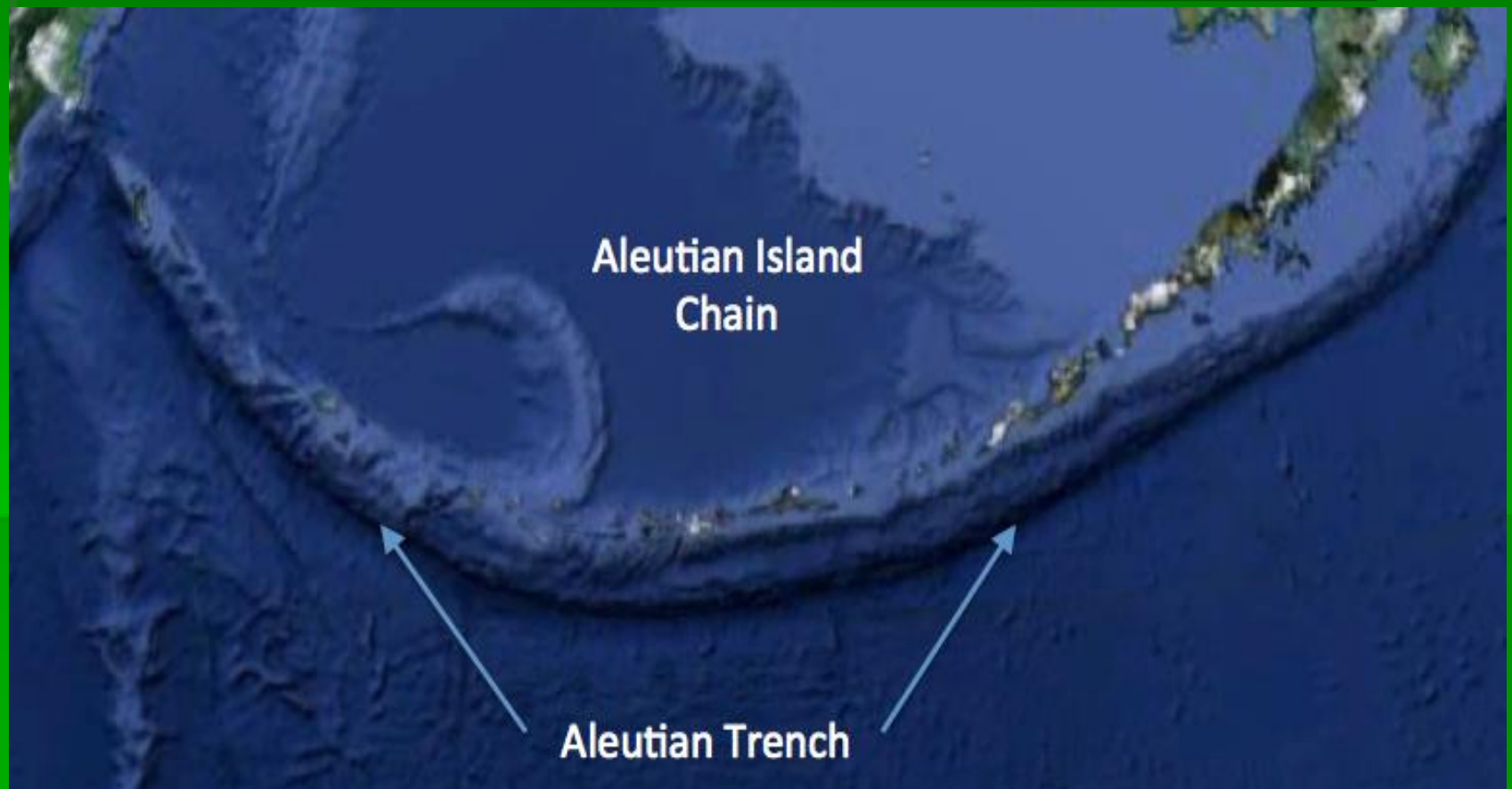


Earthquake hypocenter ranges (orange 0-33, yellow 33-70, green 70-150, blue 150-300, pink 300-500, red 500-800)

Ocean – continent collision



Location of Aleutian trench and island chain



Subduction model- Aleutian plate boundary

